coursera | Macmillan Learning

Advanced: Eigenvectors and Vector Spaces

Today's goal • 3/3 items

Module 1

Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

- ✓ **Eigenvalues and Eigenvectors Reading**
  Reading • 2h
- Eigenvalues and Eigenvectors Practice Assignment
  Practice Assignment • 20 min
- Eigenvalues and Eigenvectors Graded Assignment
  Graded Assignment • 10 min

Diagonalization

- Diagonalization Reading
  Reading • 2h
- Diagonalization Practice Assignment
  Practice Assignment • 16 min
- Diagonalization Graded Assignment
  Graded Assignment • 10 min

Complex Eigenvalues and Eigenvectors

- Complex Eigenvalues and Eigenvectors Reading
  Reading • 1h 40m
- Complex Eigenvalues and Eigenvectors Practice Assignment
  Practice Assignment • 16 min
- Complex Eigenvalues and Eigenvectors Graded Assignment
  Graded Assignment • 8 min

Systems of Differential Equations

- Systems of Differential Equations Reading
  Reading • 1h 30m
- Systems of Differential Equations Practice Assignment
  Practice Assignment • 8 min
- Systems of Differential Equations Graded Assignment
  Graded Assignment • 8 min

Approximation Methods

- Approximation Methods Reading
  Reading • 1h 15m
- Approximation Methods Practice Assignment
  Practice Assignment • 8 min
- Approximation Methods Graded Assignment
  Graded Assignment • 10 min

Module 2

Vector Spaces

Eigenvalues and Eigenvectors Reading

2:47 / 31:50 2x

Let $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation, with $T(\mathbf{x}) = A\mathbf{x}$ for a 2 × 2 matrix A. For a given vector $\mathbf{u}$ in $\mathbf{R}^2$, we can think of the multiplication $A\mathbf{u}$ as changing the direction and length of $\mathbf{u}$. Figure 1 shows $\mathbf{u}$ and $A\mathbf{u}$ for several vectors $\mathbf{u}$ and $A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$.

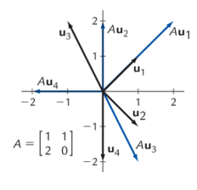

Figure 1 Plots of $\mathbf{u}$ and $A\mathbf{u}$ for the vectors $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$, and $\mathbf{u}_4$.

Note that $\mathbf{u}_1$ and $A\mathbf{u}_1$ are parallel, which means that they point in the same (or opposite) direction. This is also true of $\mathbf{u}_3$ and $A\mathbf{u}_3$.

Of particular importance in applications and in analyzing the behavior of a linear transformation are those vectors $\mathbf{u}$ such that $\mathbf{u}$ and $A\mathbf{u}$ are parallel. In Figure 1, $\mathbf{u}_1$ and $\mathbf{u}_3$ are such vectors. Algebraically, $\mathbf{u}$ and $A\mathbf{u}$ are parallel if there exists a scalar λ such that $A\mathbf{u} = \lambda\mathbf{u}$. Nonzero vectors that satisfy this equation are called *eigenvectors*.

DEFINITION 1

Let A be an $n \times n$ matrix. Then a nonzero vector $\mathbf{u}$ is an eigenvector of A if there exists a scalar λ such that

$$A\mathrm{u} = \lambda\mathrm{u} \qquad (1)$$

The scalar λ is called an eigenvalue of A.

An eigenvalue λ can be equal to zero, but an eigenvector $\mathbf{u}$ must be a nonzero vector.

When λ and $\mathbf{u}$ are related as in equation (1), we say that λ is the eigenvalue associated with $\mathbf{u}$ and that $\mathbf{u}$ is an eigenvector associated with λ.

Example 1

Let $A = \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix}$. Determine if each of

$$\mathrm{u}_1 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \quad \mathrm{u}_2 = \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \quad \text{and} \quad \mathrm{u}_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

is an eigenvector of A. For those that are, find the associated eigenvalue.

Solution

Starting with $\mathbf{u}_1$, we have

$$A\mathrm{u}_1 = \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 35 \\ 28 \end{bmatrix} = 7 \begin{bmatrix} 5 \\ 4 \end{bmatrix} = 7\mathrm{u}_1$$

Thus $A\mathbf{u}_1 = 7\mathbf{u}_1$, so $\mathbf{u}_1$ is an eigenvector of A with associated eigenvalue $\lambda = 7$. Calculating $A\mathbf{u}_2$, we have

$$A\mathrm{u}_2 = \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 14 \end{bmatrix}$$

Since $A\mathbf{u}_2$ is not a multiple of $\mathbf{u}_2$, this tells us that $\mathbf{u}_2$ is not an eigenvector of A. Finally,

$$A\mathrm{u}_3 = \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -2\mathrm{u}_3$$

Since $A\mathbf{u}_3 = -2\mathbf{u}_3$, it follows that $\mathbf{u}_3$ is an eigenvector of A with associated eigenvalue $\lambda = -2$. See Figure 2 for graphs of vectors.

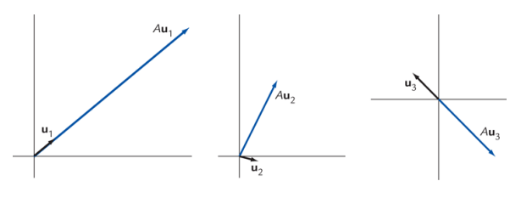

Figure 2 Graphs of vectors from Example 1. If $\mathbf{u}$ and $A\mathbf{u}$ are parallel vectors, then $\mathbf{u}$ is an eigenvector.

Referring back to $\mathbf{u}_1$ in Example 1, suppose $\mathrm{u}_4 = 3\mathrm{u}_1 = 3 \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \end{bmatrix}$. Then we have

$$A\mathrm{u}_4 = \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 15 \\ 12 \end{bmatrix} = \begin{bmatrix} 105 \\ 84 \end{bmatrix} = 7 \begin{bmatrix} 15 \\ 12 \end{bmatrix} = 7\mathrm{u}_4$$

Therefore $\mathbf{u}_4 = 3\mathbf{u}_1$ is also an eigenvector of A associated with $\lambda = 7$. We could have used any nonzero scalar in place of 3 and achieved the same result, so any nonzero multiple of $\mathbf{u}_1$ is also an eigenvector of A associated with $\lambda = 7$. Theorem 2 generalizes this observation.

THEOREM 2

Let A be a square matrix, and suppose that $\mathbf{u}$ is an eigenvector of A associated with eigenvalue λ. Then for any scalar $c \neq 0$, $c\mathbf{u}$ is also an eigenvector of A associated with λ.

We require $c \neq 0$ because eigenvectors must be nonzero.

Proof

Let $\mathbf{u}$ be an eigenvector of A with associated eigenvalue λ. Then for any scalar $c \neq 0$, we have

$A(c\mathbf{u}) = c(A\mathbf{u}) = c(\lambda\mathbf{u}) = \lambda(c\mathbf{u})$

so that $c\mathbf{u}$ is also an eigenvector of A associated with eigenvalue λ.

We address the problem of finding the eigenvalues shortly.

Finding Eigenvectors

Here we consider the problem of finding the eigenvectors associated with a known eigenvalue. Let's start with an example.

Example 2

Take as known that $\lambda = 3$ and $\lambda = 2$ are eigenvalues for

$$A = \begin{bmatrix} 4 & 4 & -2 \\ 1 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix}$$

Find the eigenvectors associated with each eigenvalue.

Solution

Starting with $\lambda = 3$, we need to find all nonzero vectors $\mathbf{u}$ such that $A\mathbf{u} = 3\mathbf{u}$. Since $3\mathbf{u} = 3I_3\mathbf{u}$, our equation becomes

$$A\mathrm{u} = 3I_3\mathrm{u} \quad \Rightarrow \quad A\mathrm{u} - 3I_3\mathrm{u} = 0 \quad \Rightarrow \quad (A - 3I_3)\,\mathrm{u} = 0$$

Thus we need to find the solutions to the homogeneous system with coefficient matrix

$$A - 3I_3 = \begin{bmatrix} 4 & 4 & -2 \\ 1 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 4 & -2 \\ 1 & 1 & -1 \\ 3 & 6 & -4 \end{bmatrix}$$

Forming an augmented matrix and performing the indicated row operations, we have

$$\left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 1 & 1 & -1 & 0 \\ 3 & 6 & -4 & 0 \end{array}\right] \begin{array}{c} -R_1 + R_2 \to R_2 \\ -3R_1 + R_3 \to R_3 \\ -2R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

After back substitution, we find that the system $(A - 3I_3)\mathbf{u} = \mathbf{0}$ has general solution

$$\mathrm{u} = s \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

We can verify that $\mathbf{u}$ is an eigenvector associated with $\lambda = 3$ by computing

$$A\mathrm{u} = \begin{bmatrix} 4 & 4 & -2 \\ 1 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix} \left(s \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right) = s \begin{bmatrix} 4 & 4 & -2 \\ 1 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = s \begin{bmatrix} 6 \\ 3 \\ 9 \end{bmatrix} = 3\mathrm{u}$$

The procedure is the same for $\lambda = 2$. This time we need to find the general solution to the homogeneous system $(A - 2I_3)\mathbf{u} = \mathbf{0}$, where

$$A - 2I_3 = \begin{bmatrix} 4 & 4 & -2 \\ 1 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 \\ 1 & 2 & -1 \\ 3 & 6 & -3 \end{bmatrix}$$

The augmented matrix and corresponding echelon form are

$$\left[\begin{array}{ccc|c} 2 & 4 & -2 & 0 \\ 1 & 2 & -1 & 0 \\ 3 & 6 & -3 & 0 \end{array}\right] \begin{array}{c} R_1 \leftrightarrow R_2 \\ -2R_1 + R_2 \to R_2 \\ -3R_1 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

After back substitution, we find that this system has general solution

$$\mathrm{u} = s_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad (2)$$

As long as s_1 or s_2 (or both) is nonzero, then $\mathbf{u}$ will be an eigenvector associated with $\lambda = 2$. We can check by computing

$$\begin{aligned} A\mathrm{u} &= A\left(s_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) = s_1 A \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s_2 A \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ &= s_1 \begin{bmatrix} -4 \\ 2 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = 2\left(s_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) = 2\mathrm{u} \end{aligned}$$

Recall that a set of vectors is a subspace if it contains $\mathbf{0}$, is closed under addition, and is closed under scalar multiplication.

If the zero vector is included, the eigenvectors in Example 2 associated with each eigenvalue form a subspace. This is always true of the set of eigenvectors associated with a given eigenvalue.

THEOREM 3

Let A be an $n \times n$ matrix with eigenvalue λ. Let S denote the set of all eigenvectors associated with λ, together with the zero vector $\mathbf{0}$. Then S is a subspace of $\mathbf{R}^n$.

We can also think of S as the null space of the matrix $A - \lambda I_n$.

Proof

We show S is a subspace by verifying the three required conditions for a subspace. First, by definition $\mathbf{0}$ is in S. Second, Theorem 2 tells us that if $\mathbf{u}$ is an eigenvector associated with λ, then so is $c\mathbf{u}$ for $c \neq 0$. If $c = 0$, then $c\mathbf{u} = \mathbf{0}$ is in S, so S is closed under scalar multiplication. Third, if $\mathbf{u}_1$ and $\mathbf{u}_2$ are both eigenvectors associated with λ, then

$$A\,(\mathrm{u}_1 + \mathrm{u}_2) = A\mathrm{u}_1 + A\mathrm{u}_2 = \lambda\mathrm{u}_1 + \lambda\mathrm{u}_2 = \lambda\,(\mathrm{u}_1 + \mathrm{u}_2)$$

so that $\mathbf{u}_1 + \mathbf{u}_2$ is also an eigenvector associated with λ. Therefore S is closed under addition, and so S is a subspace.

DEFINITION 4

Let A be a square matrix with eigenvalue λ. The subspace of all eigenvectors associated with λ, together with the zero vector, is called the eigenspace of λ.

Each distinct eigenvalue has its own associated eigenspace.

For instance, in Example 2 we see from (2) that the set

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

forms a basis for the eigenspace of $\lambda = 2$.

Finding Eigenvalues

Let's review what we have learned so far. If we know an eigenvalue λ for a given $n \times n$ matrix A, then we can find the associated eigenvectors by solving the linear system $A\mathbf{u} = \lambda\mathbf{u}$, or equivalently, the homogeneous system

$$(A - \lambda I_n)\,\mathrm{u} = 0 \qquad (3)$$

If we know the eigenvalue, then this is a problem that we know how to solve. Finding the eigenvalues is a different problem that we have not yet considered. The next theorem shows how to use determinants to find eigenvalues.

THEOREM 5

Let A be an $n \times n$ matrix. Then λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof

λ is an eigenvalue of A if and only if there exists a nontrivial solution to $A\mathbf{u} = \lambda\mathbf{u}$. This is equivalent to the existence of a nontrivial solution to (3), which by the Unifying Theorem, Version 7 is true if and only if $\det(A - \lambda I_n) = 0$.

Example 3

Find the eigenvalues for $A = \begin{bmatrix} 3 & 3 \\ 6 & -4 \end{bmatrix}$.

Solution

Our strategy is to determine the values of λ that satisfy $\det(A - \lambda I_2) = 0$. We have

$$A - \lambda I_2 = \begin{bmatrix} 3 & 3 \\ 6 & -4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} (3-\lambda) & 3 \\ 6 & (4-\lambda) \end{bmatrix}$$

Next, we compute the determinant,

$$\det\,(A - \lambda I_2) = (3 - \lambda)\,(-4 - \lambda) - 18 = \lambda^2 + \lambda - 30$$

Setting $\det(A - \lambda I) = 0$, we have

$$\lambda^2 + \lambda - 30 = 0 \quad \Rightarrow \quad (\lambda - 5)\,(\lambda + 6) = 0 \quad \Rightarrow \quad \lambda = 5 \text{ or } \lambda = -6$$

Thus the eigenvalues for A are $\lambda = 5$ and $\lambda = -6$.

The polynomial that we get from $\det(A - \lambda I)$ is called the characteristic polynomial of A, and the equation $\det(A - \lambda I) = 0$ is called the characteristic equation. The eigenvalues for a matrix A are given by the roots of the characteristic equation. In this lesson we focus on real roots; complex roots are covered in Lesson 3, Complex Eigenvalues and Eigenvectors.

Example 4

Find the eigenvalues and a basis for each eigenspace for the matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Solution

We start by finding the eigenvalues of A by computing

$$\begin{aligned} \det\,(A - \lambda I_2) &= \begin{vmatrix} (2-\lambda) & -1 \\ -1 & (2-\lambda) \end{vmatrix} \\ &= (2 - \lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 = (\lambda - 1)\,(\lambda - 3) \end{aligned}$$

Therefore the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 3$. Next, we find the eigenvectors, starting with those associated with $\lambda_1 = 1$. We find the associated eigenvectors by solving the homogeneous linear system $(A - 1 \cdot I_2)\mathbf{u} = (A - I_2)\mathbf{u} = \mathbf{0}$. Since

$$A - I_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

the augmented matrix and corresponding echelon form are

$$\left[\begin{array}{cc|c} 1 & -1 & 0 \\ -1 & 1 & 0 \end{array}\right] \begin{array}{c} R_1 + R_2 \to R_2 \\ \sim \end{array} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

Back substitution gives us

$$\text{General Solution}: \; \mathrm{u}_1 = s \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_1 = 1: \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

The eigenvectors associated with $\lambda_2 = 3$ are found by solving the homogeneous linear system $(A - 3I_2)\mathbf{u} = \mathbf{0}$. We have

$$A - 3I_2 = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$$

so the augmented matrix and corresponding echelon form are

$$\left[\begin{array}{cc|c} -1 & -1 & 0 \\ -1 & 1 & 0 \end{array}\right] \begin{array}{c} -R_1 + R_2 \to R_2 \\ \sim \end{array} \left[\begin{array}{cc|c} -1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

Back substitution gives us

$$\text{General Solution}: \; \mathrm{u}_2 = s \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_2 = 3: \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

Example 5

Find the eigenvalues and a basis for each eigenspace of

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 2 & -2 & 2 \\ 2 & 0 & 0 \end{bmatrix}$$

Solution

We determine the eigenvalues of A by calculating the characteristic polynomial,

$$\det\,(A - \lambda I_3) = \begin{vmatrix} (1-\lambda) & -3 & 3 \\ 2 & (-2-\lambda) & 2 \\ 2 & 0 & (0-\lambda) \end{vmatrix} = -\lambda^3 - \lambda^2 + 2\lambda = -\lambda\,(\lambda + 2)\,(\lambda - 1)$$

Thus the eigenvalues are $\lambda_1 = -2$, $\lambda_2 = 0$, and $\lambda_3 = 1$. Taking them in order, the eigenvectors associated with $\lambda_1 = -2$ are found by solving the homogeneous system $(A + 2I_3)\mathbf{u} = \mathbf{0}$. The augmented matrix and echelon form are

$$\left[\begin{array}{ccc|c} 3 & -3 & 3 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 2 & 0 \end{array}\right] \begin{array}{c} R_1 \leftrightarrow R_2 \\ -\frac{3}{2}R_1 + R_2 \to R_2 \\ -R_1 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 2 & 0 & 2 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Back substitution gives us

$$\text{General Solution}: \; \mathrm{u}_1 = s \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_1 = -2: \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$$

For $\lambda_2 = 0$, the homogeneous system is $(A - 0 \cdot I_2)\mathbf{u} = A\mathbf{u} = \mathbf{0}$. The augmented matrix and echelon form are

$$\left[\begin{array}{ccc|c} 1 & -3 & 3 & 0 \\ 2 & -2 & 2 & 0 \\ 2 & 0 & 2 & 0 \end{array}\right] \begin{array}{c} -2R_1 + R_1 \to R_2 \\ -R_1 + R_3 \to R_3 \\ -\frac{3}{2}R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -3 & 3 & 0 \\ 0 & 4 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This time back substitution yields

$$\text{General Solution}: \; \mathrm{u}_2 = s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_2 = 0: \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

The final eigenvalue is $\lambda_3 = 1$. The homogeneous system is $(A - I_3)\mathbf{u} = \mathbf{0}$, and the augmented matrix and echelon form are

$$\left[\begin{array}{ccc|c} 0 & -3 & 3 & 0 \\ 2 & -3 & 2 & 0 \\ 2 & 0 & -1 & 0 \end{array}\right] \begin{array}{c} R_1 \leftrightarrow R_3 \\ -R_1 + R_2 \to R_2 \\ -R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 2 & 0 & -1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

With back substitution, we find

$$\text{General Solution}: \; \mathrm{u}_3 = s \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_3 = 1: \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$$

Example 6

Find the eigenvalues and a basis for each eigenspace of

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 0 & 1 \\ -1 & -2 & 3 \end{bmatrix}$$

Solution

We start out by finding the eigenvalues for A by computing

$$\det\,(A - \lambda I_3) = \begin{vmatrix} (1-\lambda) & -2 & 1 \\ -1 & -\lambda & 1 \\ -1 & -2 & (3-\lambda) \end{vmatrix} = -\lambda^3 + 4\lambda^2 - 4\lambda = -\lambda\,(\lambda - 2)^2$$

From the factored form we see that our matrix has two distinct eigenvalues, $\lambda_1 = 0$ and $\lambda_2 = 2$.

Now we find the eigenvectors. Starting with those associated with $\lambda_1 = 0$, we solve the homogeneous system $(A - 0 \cdot I_3)\mathbf{u} = A\mathbf{u} = \mathbf{0}$. The augmented matrix and echelon form are

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right] \begin{array}{c} R + R_2 \to R_3 \\ R_1 + R_3 \to R_3 \\ -2R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Back substitution produces

$$\text{General Solution}: \; \mathrm{u}_1 = s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_1 = 0: \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

For the eigenvalue $\lambda_2 = 2$ we form the augmented matrix for the system $(A - 2I_3)\mathbf{u} = \mathbf{0}$ and reduce to echelon form,

$$\left[\begin{array}{ccc|c} -1 & -2 & 1 & 0 \\ -1 & -2 & 1 & 0 \\ -1 & -2 & 1 & 0 \end{array}\right] \begin{array}{c} -R_1 + R_2 \to R_2 \\ -R_1 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This time back substitution produces

$$\begin{aligned} \text{General Solution}: \; \mathrm{u}_2 &= s_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \\ \Rightarrow \text{Basis for Eigenspace of } \lambda_2 &= 2: \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\} \end{aligned}$$

Multiplicities

In Example 6 the factored form of the characteristic polynomial is

$$-\lambda\,(\lambda - 2)^2 = -\,(\lambda - 0)^1\,(\lambda - 2)^2$$

The *multiplicity* of an eigenvalue is equal to its factor's exponent. In this case, we say that $\lambda = 0$ has multiplicity 1 and $\lambda = 2$ has multiplicity 2.

In general, for a polynomial $P(x)$, a root a of $P(x) = 0$ has multiplicity if $P(x) = (x - a)^r Q(x)$ with $Q(a) \neq 0$. When discussing eigenvalues, the phrase "λ has multiplicity r" means that λ is a root of the characteristic polynomial with multiplicity r.

Reviewing our previous examples, we see that the dimension of the eigenspaces matched the multiplicities of the associated eigenvalues. This happens most of the time, but not always.

Put another way, the multiplicity is the number of times a root is repeated.

Example 7

Find the eigenvalues and a basis for each eigenspace of

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -1 & 0 \\ 1 & -2 & 0 \end{bmatrix}$$

Solution

The characteristic polynomial of A is

$$\det\,(A - \lambda I_3) = \begin{vmatrix} -\lambda & 2 & -1 \\ 1 & (-1-\lambda) & 0 \\ 1 & -2 & -\lambda \end{vmatrix} = -\lambda^3 - \lambda^2 + \lambda + 1 = -\,(\lambda - 1)\,(\lambda + 1)^2$$

Thus A has two distinct eigenvalues, $\lambda_1 = -1$ (multiplicity 2) and $\lambda_2 = 1$ (multiplicity 1).

To find the eigenvectors associated with $\lambda_1 = -1$, we solve the homogeneous system $(A + I_3)\mathbf{u} = \mathbf{0}$. The augmented matrix and echelon form are

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 \end{array}\right] \begin{array}{c} -R_1 + R_2 \to R_2 \\ -R_1 + R_3 \to R_3 \\ -2R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Back substitution produces

$$\text{General Solution}: \; \mathrm{u}_1 = s \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_1 = -1: \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Note that although the eigenvalue $\lambda_1 = -1$ has multiplicity 2, the eigenspace has dimension 1.

For $\lambda_2 = 1$, the augmented matrix for the system $(A - I_3)\mathbf{u} = \mathbf{0}$ and echelon form are

$$\left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 1 & -2 & 0 & 0 \\ 1 & -2 & -1 & 0 \end{array}\right] \begin{array}{c} R_1 + R_2 \to R_2 \\ R_1 + R_3 \to R_3 \\ -2R_2 + R_3 \to R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Back substitution gives us

$$\text{General Solution}: \; \mathrm{u}_2 = s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \Rightarrow \quad \text{Basis for Eigenspace of } \lambda_2 = 1: \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

As demonstrated in Example 7, it is possible for the dimension of an eigenspace to be less than the multiplicity of the associated eigenvalue. However, the opposite cannot happen.

THEOREM 6

Let A be a square matrix with eigenvalue λ. Then the dimension of the associated eigenspace is less than or equal to the multiplicity of λ.

The proof is beyond the scope of this book and is omitted.

The methods developed in this lesson work well for small matrices, but they can be impractical for large complicated matrices. For instance, suppose

$$A = \begin{bmatrix} 245 & -254 & -252 & -46 & -224 \\ 161 & -168 & -174 & -32 & -148 \\ -39 & 40 & 45 & 7 & 38 \\ 27 & -28 & -32 & -6 & -26 \\ 110 & -113 & -110 & -21 & -101 \end{bmatrix} \qquad (4)$$

Computing the determinant by hand to find the characteristic polynomial for A is not easy. Using computer software, we find that

$$\det\,(A - \lambda I_5) = -\lambda^5 + 15\lambda^4 - 3\lambda^3 - 287\lambda^2 - 192\lambda + 468$$

This polynomial is challenging to factor. In fact, there is no general algorithm for factoring polynomials of degree 5 or more. (Not even on a computer.) Thus we cannot find the eigenvalues and eigenvectors using our existing methods.

Numerous applications require eigenvalues and eigenvectors from large matrices. In principle, we could use numerical methods to find approximations to the eigenvalues, and use these to find the eigenvectors. However, for various reasons this does not work well in practice. Instead, there are algorithms that lead directly to approximations to the eigenvectors, bypassing the need to first find the eigenvalues. These are described in Lesson 5, Approximation Methods, where we analyze the matrix A in (4).

The Unifying Theorem, Version 8

Although $\mathbf{u} = \mathbf{0}$ is not allowed as an eigenvector, it is fine to have $\lambda = 0$ as an eigenvalue. From Theorem 5, we know that $\lambda = 0$ is an eigenvalue of an $n \times n$ matrix A if and only if $\det(A - 0I_n) = \det(A) = 0$. Put another way, $\lambda = 0$ is *not* an eigenvalue of A if and only if $\det(A) \neq 0$. This observation provides us with another condition for the Unifying Theorem.

THEOREM 7

(THE UNIFYING THEOREM, VERSION 8) Let $\mathcal{S} = \{a_1, \ldots, a_n\}$ be a set of n vectors in $\mathbf{R}^n$, let $A = [\mathbf{a}_1 \ldots \mathbf{a}_n]$, and let $T: \mathbf{R}^n \rightarrow \mathbf{R}^n$ be given by $T(\mathbf{x}) = A\mathbf{x}$. Then the following are equivalent:

1. $\mathcal{S}$ spans $\mathbf{R}^n$.
2. $\mathcal{S}$ is linearly independent.
3. $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$ in $\mathbf{R}^n$.
4. T is onto.
5. T is one-to-one.
6. A is invertible.
7. $\ker(T) = \{\mathbf{0}\}$.
8. $\mathcal{S}$ is a basis for $\mathbf{R}^n$.
9. $\mathrm{col}(A) = \mathbf{R}^n$.
10. $\mathrm{row}(A) = \mathbf{R}^n$.
11. $\mathrm{rank}(A) = n$.
12. $\det(A) \neq 0$.
13. $\lambda = 0$ is not an eigenvalue of A.

Proof

From the Unifying Theorem, Version 7, we know that (1) through (12) are equivalent. The comments above show that (12) and (13) are equivalent, and so it follows that all 13 conditions are equivalent.

Example 8

Show that $\lambda = 0$ is an eigenvalue for the matrix

$$A = \begin{bmatrix} 3 & -1 & 5 \\ 2 & 1 & 0 \\ 4 & 1 & 2 \end{bmatrix}$$

Solution

From the Unifying Theorem, Version 8, $\lambda = 0$ is an eigenvalue of A if and only if $\det(A) = 0$. By the Shortcut Method, we have

$$\det(A) = (6 + 0 + 10) - (20 - 4 + 0) = 0$$

so $\lambda = 0$ is an eigenvalue of A.

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